

SOP of DIA-NN v2.2.0 on HPC

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Introduction

Amongst the data acquisition techniques employed in mass spectrometry, the two most popular ones are data-independent acquisition (DIA) and data-dependent acquisition (DDA). DIA is the superior method for analyzing complex proteomic samples due to its numerous advantages over DDA, including its ability to detect low-abundance peptides, its increased specificity, and its greater reproducibility. With its comprehensive and accurate analysis, DIA has become an indispensable tool for researchers in the fields of drug discovery and clinical research, helping to advance our understanding of complex biological systems and diseases. However, the data analysis of DIA is much more complex than DDA. DIA-NN and Spectronaut are the most popular DIA analysis tools. This SOP focuses on the operation procedure of DIA-NN and JUMP Shiny to generate the DIA report.

Summary of scripts

```
bsub -P DIANN -n 64 -q priority -R "rusage[mem=4G]" -R "span[hosts=1]" -Is bash
cd
mkdir DIATutorial_DIANN
cp -R /hpcf/authorized_apps/proteomics_apps/DIANN_CMD/CMD2 ./DIATutorial_DIANN
cd ./DIATutorial_DIANN/CMD2
# edit d_DIANNParamsOption.params
bash cmd_DIANNParams.sh
bash workflow.sh
```

Procedure

This part includes the operation procedure of DIA-NN and JUMP Shiny.

1. Apply a computing node

In your home path on the login node, e.g. [zyuan1@splprhpc12 ~]\$, apply a computing node on HPC with the following command:

```
bsub -P DIANN -n 64 -q priority -R "rusage[mem=4G]" -R "span[hosts=1]" -Is bash
```

Note: It may take a long waiting time to get a computing node on a specific queue. You can try 4 queues: priority, standard, large_mem, large_core_count to check which is faster. Try this command to check the status for each queue:

```
qstat -l standard; qstat -l priority; qstat -l large_mem; qstat -l large_core_count
```

2. Generate parameters and workflow of DIA-NN

Here is the test data for the following procedure, which includes the data path, the fasta file, and commands to generate DIA-NN parameters and workflow as follows. (Note: The original data path and the result path can be different.)

/home/zyuan1/DIATutorial_DIANN2/data1

```
/research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN2/data1/
Name
..
hela_200ngDIA_test2_S1-B2_1_299.d
hela_200ngDIA_test3_S1-B3_1_300.d
hela_200ngDIA_test4_S1-B4_1_301.d
hela_200ngDIA_test5_S1-B5_1_302.d
```

/home/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN

```
/research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN/
Name
..
human_TryP_mc2_c0_lib
human_TryP_mc2_c0_prot_anno
diann_1_LibGen.params
HUMAN_6_sp_tr_uc_snp_con_cu.fasta
```

/hpcf/authorized_apps/proteomics_apps/DIANN_CMD/CMD2

```
/research/rgs01/applications/hpcf/authorized_apps/proteomics_apps/DIANN_CMD/CMD2/
Name
..
cmd_DIANNParams.sh
d_DIANNParamsOption.params
```

Here are the operation steps.

2.1. Copy commands to generate DIA-NN parameters from above path to your path (e.g. /home/zyuan1):

```
cd
```

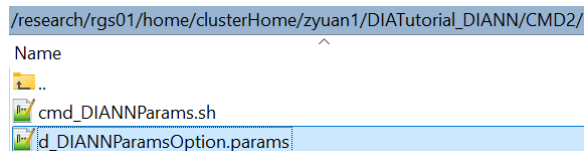
```
mkdir DIATutorial_DIANN
```

```
cp -R /hpcf/authorized_apps/proteomics_apps/DIANN_CMD/CMD2 ./DIATutorial_DIANN
```

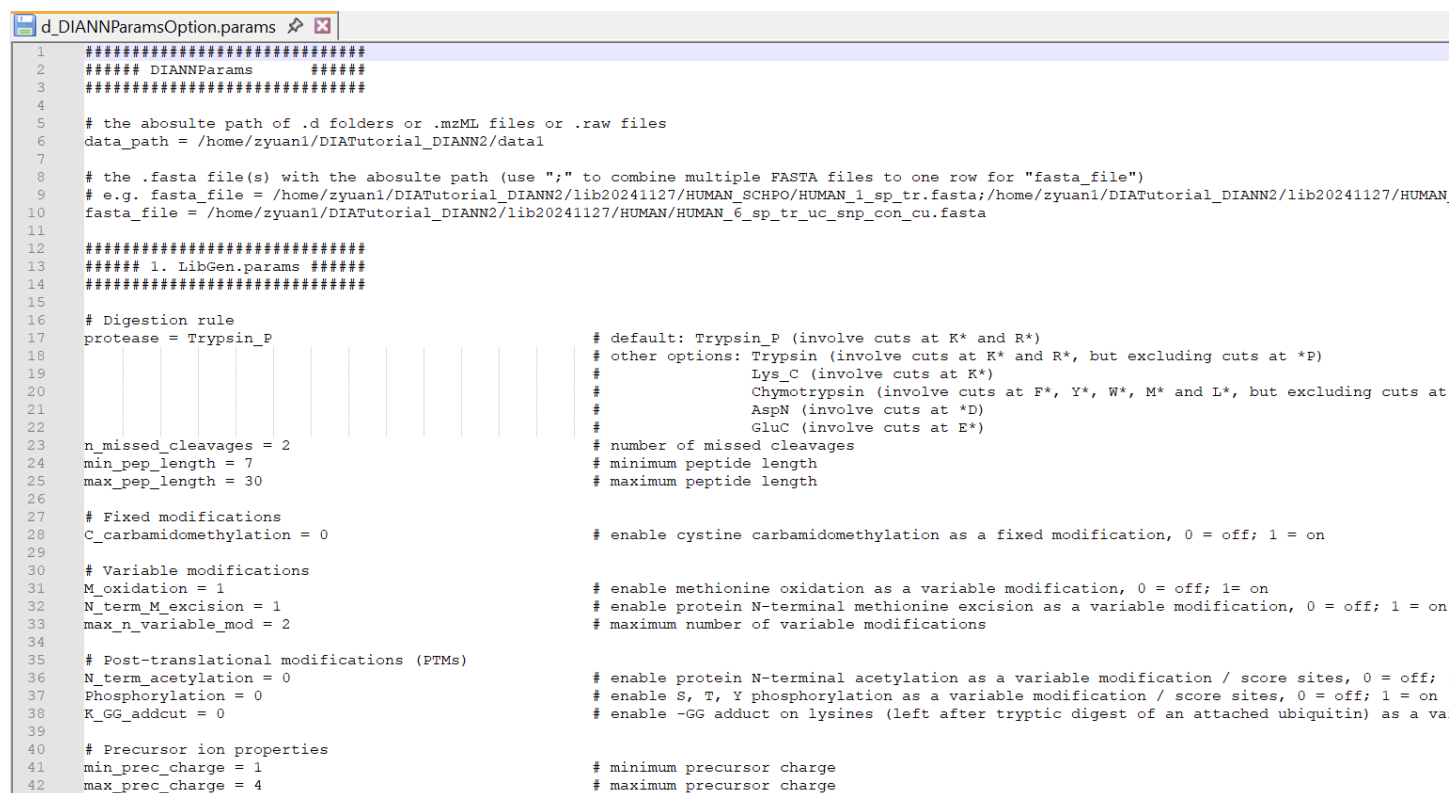
2.2. Open 'd_DIANNParamsOption.params' in your path, and change 'data_path', 'fasta_file' and other parameters to yours (e.g. **additional_options** = --var-mod Dimethyl,28.0313,KR is to search Dimethyl)

Note: these parameters are the same as the GUI parameters of DIANN v2.2.0. Change 'd_DIANNParamsOption.params', and other *.params files will be automatically changed (diann_1_LibGen.params, diann_2_Search.params, diann_3_iqQuan.params).

/home/zyuan1/DIATutorial_DIANN/CMD2



/home/zyuan1/DIATutorial_DIANN/CMD2/d_DIANNParamsOption.params



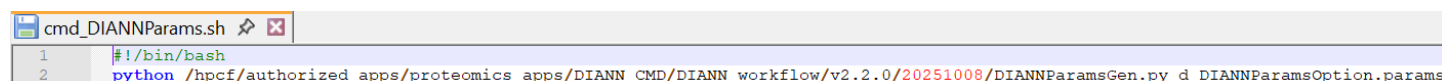
2.3. Run the following commands to generate DIA-NN parameters and workflow

```
cd
```

```
cd ./DIATutorial_DIANN/CMD2
```

```
bash cmd_DIANNParams.sh
```

/home/zyuan1/DIATutorial_DIANN/CMD2/cmd_DIANNParams.sh



Output of the command running (highlighted):

/home/zyuan1/DIATutorial_DIANN/CMD2

```
/research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/CMD2/
Name
└─ ..
└─ cmd_DIANNParams.sh
└─ d_DIANNParamsOption.params
└─ diann_1_LibGen.params
└─ diann_2_Search.params
└─ diann_3_iqQuan.params
└─ workflow.sh
```

3. Run the workflow of DIA-NN

Running DIA-NN includes 3 steps: predicting the library, searching and quantifying precursors, quantifying proteins. If you change 'd_DIANNParamsOption.params' in step 2.2, and the 3 *.params files will be automatically changed: diann_1_LibGen.params, diann_2_Search.params, diann_3_iqQuan.params (Appendix A, B, C).

Run the following commands to run DIA-NN workflow with 3 steps: predicting the library, searching and quantifying precursors, quantifying proteins.

```
cd
```

```
cd ./DIATutorial_DIANN/CMD2
```

```
bash workflow.sh
```

/home/zyuan1/DIATutorial_DIANN/CMD2/workflow.sh

```
workflow.sh
1 #!/bin/bash
2 module purge
3 # activate the 'diann_iq_cmd' Conda environment
4 module load gcc/13.1.0
5 module load conda3/202402
6 conda activate /hpcf/authorized_apps/teomics_apps/conda/teoda2
7
8 # make softlink of the data_path
9 mkdir -p /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data
10 ln -s /home/zyuan1/DIATutorial_DIANN2/data/* /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data
11
12 # copy parameter files
13 cp /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/CMD2/diann_1_LibGen.params /home/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN
14 cp /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/CMD2/diann_2_Search.params /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data
15 cp /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/CMD2/diann_3_iqQuan.params /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data
16
17 # predict the library (skipped as they already exist)
18 cd /home/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN
19 /hpcf/authorized_apps/teomics_apps/conda/teoda2/bin/python /hpcf/authorized_apps/teomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/diann_LibGen.py diann_1_LibGen.params
20 /hpcf/authorized_apps/teomics_apps/conda/teoda2/bin/python /hpcf/authorized_apps/teomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/parse_FASTA.py human_TryP_mc2_c0_prot
21
22 # search and quantify precursors, quantify proteins
23 cd /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data
24 /hpcf/authorized_apps/teomics_apps/conda/teoda2/bin/python /hpcf/authorized_apps/teomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/diann_Search.py diann_2_Search.params
25 /hpcf/authorized_apps/teomics_apps/conda/teoda2/bin/python /hpcf/authorized_apps/teomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/diann_iqQuan.py diann_3_iqQuan.params
26
27 # deactivate the 'diann_iq_cmd' Conda environment after the task is complete
28 conda deactivate
```

Note: If the library has been generated, in workflow.sh, the library generation step will be automatically skipped by commenting the corresponding commands.

```
1 #!/bin/bash
2 module purge
3 # activate the 'diann_iq_cmd' Conda environment
4 module load gcc/13.1.0
5 module load conda3/202402
6 conda activate /hpcf/authorized_apps/teomics_apps/conda/teoda2
7
8 # make softlink of the data_path
9 mkdir -p /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data
10 ln -s /home/zyuan1/DIATutorial_DIANN2/data/* /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data
11
12 # copy parameter files
13 #cp /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/CMD2/diann_1_LibGen.params /home/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN
14 #cp /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/CMD2/diann_2_Search.params /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data
15 #cp /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/CMD2/diann_3_iqQuan.params /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data
16
17 # predict the library (skipped as they already exist)
18 #cd /home/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN
19 #/hpcf/authorized_apps/teomics_apps/conda/teoda2/bin/python /hpcf/authorized_apps/teomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/diann_LibGen.py diann_1_LibGen.param
20 #/hpcf/authorized_apps/teomics_apps/conda/teoda2/bin/python /hpcf/authorized_apps/teomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/parse_FASTA.py human_TryP_mc2_c0_pro
21
22 # search and quantify precursors, quantify proteins
23 cd /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data
24 /hpcf/authorized_apps/teomics_apps/conda/teoda2/bin/python /hpcf/authorized_apps/teomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/diann_Search.py diann_2_Search.params
25 /hpcf/authorized_apps/teomics_apps/conda/teoda2/bin/python /hpcf/authorized_apps/teomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/diann_iqQuan.py diann_3_iqQuan.params
26
27 # deactivate the 'diann_iq_cmd' Conda environment after the task is complete
28 conda deactivate
```

Output of the command running (highlighted):

/home/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN

/research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN/

Name

..

human_TryP_mc2_c0.lib

human_TryP_mc2_c0_prot_anno

diann_1_LibGen.params

HUMAN_6_sp_tr_uc_snp_con_cu.fasta

/home/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN/human_TryP_mc2_c0.lib

/research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN/human_TryP_mc2_c0.lib/	
Name	Size
diann_LibGen.params	4 KB
human_TryP_mc2_c0.lib.log.txt	3 KB
human_TryP_mc2_c0.lib.predicted.speclib	2,579,333 KB

/home/zyuan1/DIATutorial_DIANN/data

/research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data/

Name

..

hela_200ngDIA_test2_S1-B2_1_299.d

hela_200ngDIA_test3_S1-B3_1_300.d

hela_200ngDIA_test4_S1-B4_1_301.d

hela_200ngDIA_test5_S1-B5_1_302.d

quan_results_20251003

search_results_20251003

diann_2_Search.params

diann_3_iqQuan.params

hela_200ngDIA_test2_S1-B2_1_299.d.quant

hela_200ngDIA_test3_S1-B3_1_300.d.quant

hela_200ngDIA_test4_S1-B4_1_301.d.quant

hela_200ngDIA_test5_S1-B5_1_302.d.quant

/home/zyuan1/DIATutorial_DIANN/data/search_results_20251003

/research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data/search_results_20251003/

Name

^

..

report_xic

diann_Search.params

report.gg_matrix.tsv

report.log.txt

report.manifest.txt

report.parquet

report.pg_matrix.tsv

report.pr_matrix.tsv

report.protein_description.tsv

report.site_report.parquet

report.stats.tsv

report.UniMod_35_sites_90.tsv

report.UniMod_35_sites_99.tsv

report.unique_genes_matrix.tsv

report-first-pass.manifest.txt

report-first-pass.parquet

report-first-pass.pr_matrix.tsv

report-first-pass.site_report.parquet

report-first-pass.stats.tsv

/home/zyuan1/DIATutorial_DIANN/data/quant_results_20251003/publication

/research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data/quant_results_20251003/publication/

Name

^

..

uni_prot_quant.csv

uni_prot_quant_rmNA.csv

uni_prot_quant_rmNA_norm.csv

uni_prot_quant_rmNA_norm_rawintensity.csv

uni_prot_quant_rmNA_norm_Rshiny.csv

uni_prot_quant_rmNA_rawintensity.csv

uni_prot_quant_rmNA_Rshiny.csv

4. Prepare JUMP Shiny input with DIA-NN output

Because of the name issue (in JUMP Shiny, ‘_’ is acceptable, but ‘-’ is not; name starting with a number is not acceptable.), the above ‘uni_prot_quant_rmNA_norm_Rshiny.csv’ can be changed.

Here are the operation steps.

- 4.1. Download above ‘uni_prot_quant_rmNA_norm_Rshiny.csv’ from HPC to your PC.
- 4.2. Make the sample name shorter (e.g. ‘hela_200ngDIA_test5_S1_B5_1_302’ -> ‘hela_200ngDIA_test5’)

Original sample name

	A	B	C	D	E	F	G	H	I	
1	Accession	Genes	Protein	De	hela_200n	hela_200n	hela_200n	hela_200ngDIA_test5_S1_B5_1_302		
2	tr A0A087	GPR89B	G protein-c	18349.35	18029.67	19043.89	17289.56			
3	tr A0A096	AAK1	AP2 assoc	9741.319	10473.39	8715.878	9115.79			
4	tr A0A096	ATP6	ATP synth	291348.2	283614.1	292135	301286			
	<	>	uni_prot_quant_rmNA_norm_Rshiny		+					

Shorter sample name

	A	B	C	D	E	F	G	H
1	Accession	Genes	Protein.De	hela_200n	hela_200n	hela_200n	hela_200ngDIA_test5	
2	tr A0A087	GPR89B	G protein-c	18349.35	18029.67	19043.89	17289.56	
3	tr A0A096	AAK1	AP2 assoc	9741.319	10473.39	8715.878	9115.79	
4	tr A0A096	ATP6	ATP synth	291348.2	283614.1	292135	301286	
< >		uni_prot_quan_rmNA_norm_Rshiny				+		

4.3. Prepare the 'sample_info.xls' file

First, create an empty .txt file 'sample_info.txt'. Second, change it to 'sample_info.xls'. Third, put the following information in 'sample_info.xls'.

Sample	Group
hela_200ngDIA_test2	Group1
hela_200ngDIA_test3	Group1
hela_200ngDIA_test4	Group2
hela_200ngDIA_test5	Group2

	A	B
1	Sample	Group
2	hela_200ngDIA_test2	Group1
3	hela_200ngDIA_test3	Group1
4	hela_200ngDIA_test4	Group2
5	hela_200ngDIA_test5	Group2
< >		sample_info

5. Run JUMP Shiny

In Web Browser (e.g. Chrome), input the URL: <http://svlprshiny01.stjude.org/JUMPrshiny/>

You can refer to the Tutorial tab in the left sidebar to learn how to do 'Exploratory Analysis' and 'Differential Expression'. Here is a brief introduction to run JUMP Shiny.

5.1. Click on the Exploratory Analysis tab in the left sidebar

5.2. Upload data with 'uni_prot_quan_rmNA_Rshiny.csv', and upload sample information with 'sample_info.xls'

5.3. Click on the 'Assign group information' button, and check 'Intensity Distribution' and 'PCA' for data quality

5.4. Click on the Differential Expression tab in the left sidebar

5.5. Click on the 'Run Differential Expression Test' button for pairwise comparison of Group1 and Group2, click on the 'Download Result Table' button to obtain the result table with p-value, FDR, Log2Fold for all proteins

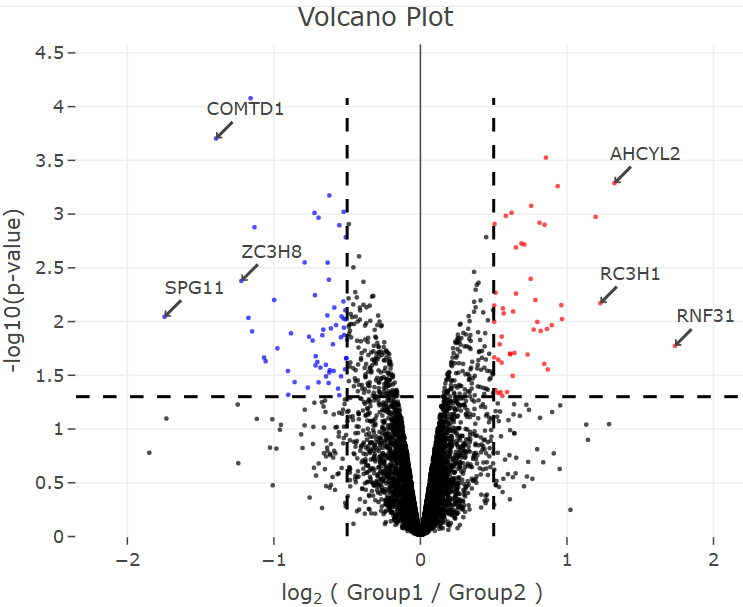
DEresult.csv

	A	B	C	D	E	F	G	H	I	J	K
1		GN	Description	hela_200n	hela_200n	hela_200n	hela_200n	p-value	FDR	Log2Fold(Group1/Group2)	
2	A0A087W	GPR89B	G protein-c	14.16344	14.13809	14.21704	14.07761	0.96573	0.995006	0.003437	
3	A0A096LP	AAK1	AP2 assoc	13.2499	13.35444	13.08943	13.15415	0.06773	0.728289	0.18038	
4	A0A096VT	ATP6	ATP synth	18.15238	18.11357	18.15628	18.20077	0.476052	0.915436	-0.04555	

5.6. Click on the ‘SD within group’ page at the bottom, click on the ‘Calculate SD of Log2FC’ button for Group1 and Group2, and record 2SD as the Log2Fold cutoff (i.e. 0.5)

5.7. Click on the ‘Volcano Plot’ page, set p-value<0.05 and |Log2-fold|>0.5, click on the ‘Run Volcano Plot’ button to generate the volcano plot, click on the ‘Download DE Table’ button to obtain the DE table, open the DE table, sort by Log2Fold, put top 3 down-regulated protein IDs and top 3 up-regulated protein IDs in the box of ‘Interested protein to label on plot’ under ‘Advanced Parameters’, and the volcano plot with labels is ready to download.

Volcano Plot.svg



DEresult.csv and Volcano Plot.svg can be sent to users.

Appendix

Appendix shows the details to set DIA-NN parameters for 3 steps: predicting the library, searching and quantifying precursors, quantifying proteins. For more details of DIA-NN parameters, please refer to [DIA-NN manual](https://github.com/vdemichev/diann) (<https://github.com/vdemichev/diann>).

d_DIANNParamsOption.params

Note: these parameters are the same as the GUI parameters of DIANN v2.2.0. Change 'd_DIANNParamsOption.params', and other *.params files will be automatically changed (diann_1_LibGen.params, diann_2_Search.params, diann_3_iqQuan.params).

```
d_DIANNParamsOption.params
5 # the absolute path of .d folders or .mzML files or .raw files
6 data_path = /home/zyuan/DIATutorial_DIANN2/data1
7
8 # the .fasta file(s) with the absolute path (use "/" to combine multiple FASTA files to one row for "fasta_file")
9 e.g. fasta_file = /home/zyuan/DIATutorial_DIANN2/lib20241127/HUMAN_SCHPO/HUMAN_1_sp_tr.fasta:/home/zyuan/DIATutorial_DIANN2/lib20241127/HUMAN_SCHPO/SCHPO_1_sp_tr.fasta
10 fasta_file = /home/zyuan/DIATutorial_DIANN2/lib20241127/HUMAN/HUMAN_6_sp_tr_us_nsp_con_cu.fasta
11
12 ##### 1. LibGen.params #####
13 #####
14 #####
15
16 # Digestion rule
17 protease = Trypsin_P # default: Trypsin_P (involve cuts at K* and R*)
18 # other options: Trypsin (involve cuts at K* and R*, but excluding cuts at *P)
19 # Lys_C (involve cuts at K*)
20 # Chymotrypsin (involve cuts at F*, Y*, W*, M* and L*, but excluding cuts at *P)
21 # AspN (involve cuts at *D)
22 # GluC (involve cuts at E*)
23 n_missed_cleavages = 2 # number of missed cleavages
24 min_pep_length = 7 # minimum peptide length
25 max_pep_length = 30 # maximum peptide length
26
27 # Fixed modifications
28 C_carbamidomethylation = 0 # enable cysteine carbamidomethylation as a fixed modification, 0 = off; 1 = on
29
30 # Variable modifications
31 M_oxidation = 1 # enable methionine oxidation as a variable modification, 0 = off; 1 = on
32 N_term_M_excision = 1 # enable protein N-terminal methionine excision as a variable modification, 0 = off; 1 = on
33 max_n_variable_mod = 2 # maximum number of variable modifications
34
35 # Post-translational modifications (PTMs)
36 N_term_acetylation = 0 # enable protein N-terminal acetylation as a variable modification / score sites, 0 = off; 1 = on
37 Phosphorylation = 0 # enable S, T, Y phosphorylation as a variable modification / score sites, 0 = off; 1 = on
38 K_GG_adduct = 0 # enable -GG adduct on lysines (left after tryptic digest of an attached ubiquitin) as a variable modification / score sites, 0 = off; 1 = on
39
40 # Precursor ion properties
41 min_prec_charge = 1 # minimum precursor charge
42 max_prec_charge = 4 # maximum precursor charge
43 min_prec_mz = 400 # minimum precursor m/z
44 max_prec_mz = 1200 # maximum precursor m/z
45
46 # Fragment ion properties
47 min_frag_mz = 200 # minimum fragment ion m/z
48 max_frag_mz = 1800 # maximum fragment ion m/z
49
50 ##### 2. Search.params #####
51 #####
52 #####
53
54 # Identification related parameters
55
56 ms2_tolerance = 0 # MS2 mass accuracy (ppm), use 0 for automatic inference
57 ms1_tolerance = 0 # MS1 mass accuracy (ppm), use 0 for automatic inference
58 n_scans_per_peak = 0 # number of scans per peak, use 0 for automatic inference
59 unrelated_runs = 0 # Different runs will be treated as unrelated, 0 = false (default); 1 = true
60 fdr_prec = 0.01 # FDR at precursor level
61 scoring = 2 # 1 = Generic (default); 2 = Peptidoforms; 3 = Proteoforms
62 machine_learning = 3 # 1 = Linear classifier; 2 = NMs (fast); 3 = NMs (cross-validated) (default)
63
64 # Match between runs (MBR)
65 MBR = 1 # create a spectral library from DIA data and reanalyze using this library, 0 = off; 1 = on (default)
66
67 # Quantification related parameters
68 quantification_strategy = 3 # 1 = Legacy (direct); 2 = QuantUMS (high accuracy); 3 = QuantUMS (high precision) (default)
69 cross_run_normalization = 4 # 1 = Global; 2 = RT-dependent (default); 3 = RT & signal-dep. (experimental); 4 = off
70
71 # Protein inference mode
72 prot_inference = 1 # 0 = off; 1 = on (default), if turned on, one protein will be assigned to only one protein group
73 proteotypicity = 4 # 1 = Isoform IDs; 2 = Protein names (from FASTA); 3 = Genes (Species-specific); 4 = Genes (default)
74
75 ##### 3. iqQuan.params #####
76 #####
77 #####
78
79 # Quantification method (Important!)
80 QuanMethod = 1 # 0 = DIANN native (use PG.TopN)
81 # 1 = MaxLFQ (use PG.MaxLFQ)
82
83 # Q value cutoffs
84 Lib.Q.Value = 0.01 # Please use the same column names from DIA-NN report
85 Lib.PG.Q.Value = 0.01
86
87 # Filter
88 allowed_miss_pct = 100 # filter out proteins based on missing value percentages
89
90 # Normalization
91 perform_normalization = 1 # 0 = off, 1 = on
92 normalization_method = median # use median or mean intensity for cross-run normalization
93 percentage_trimmed = 15 # percentage of most variable intensities to be trimmed for the normalization
94 norm_species = ALL # if it is on, normalized to : ALL (default), 1 species for Benchmark: HUMAN; MOUSE; RAT; YEAST; SCHPO; ECOLI; DROME; ARATH; CAEEL; CHICK; DANRE
95
96 ##### Other params #####
97 #####
98
99 library_generation = 2 # 1 = IDs profiling; 2 = IDs, RT & IM profiling (default); 3 = Smart profiling; 4 = Full profiling
100 speed_RAM_usage = 1 # 1 = Optimal results (default); 2 = Low RAM usage; 3 = Low RAM & high speed; 4 = Ultra-fast
101 n_CPU_threads = 64 # number of CPU threads used for running
102 log_level = 5 # specify how detailed the information reported in the log file, use numbers between 0-5
103 proteoDA2 = 0 # 0 = off, 1 = on
104 additional_options = null # null (default); others, e.g. --var-mod Dimethyl,28.0313,KR
```

Appendix A. Predicting the library: in silico spectral library generation

This step will generate a predicted spectral library based on the input FASTA file. The predicted spectral library is in .speclib format (specific to DIA-NN software) and can be used for searching both Thermo Orbitrap (.mzML) and Bruker timsTOF (.d) MS data. Run this step when you want to build a customized library.

```
diann_1_LibGen.params
1 # DIA-NN in silico spectral library generation parameters (Version: 0.1.0, Date: 08/14/2024)
2
3 # FASTA files (you can put multiple FASTA files here, separate each file by ;)
4 fasta_file = /home/zyuan/DIATutorial_DIANN2/lib20241127/HUMAN/HUMAN_6_sp_tr_uc_snp_con_cu.fasta
5
6 # Output library name
7 library_name = human_TryP_mc2_c0_lib # the library will be saved in .speclib format
8
9 # Digestion rule
10 protease = Trypsin_P # default: Trypsin_P (involve cuts at K* and R*)
11 # other options: Trypsin (involve cuts at K* and R*, but excluding cuts at *P)
12 # Lys_C (involve cuts at K*)
13 # Chymotrypsin (involve cuts at F*, Y*, W*, M* and L*, but excluding cuts at *P)
14 # AspN (involve cuts at *D)
15 # GluC (involve cuts at E*)
16 n_missed_cleavages = 2 # number of missed cleavages
17 min_pep_length = 7 # minimum peptide length
18 max_pep_length = 30 # maximum peptide length
19
20 # Fixed modifications
21 C_carbamidomethylation = 0 # enable cystine carbamidomethylation as a fixed modification, 0 = off; 1 = on
22
23 # Variable modifications
24 M_oxidation = 1 # enable methionine oxidation as a variable modification, 0 = off; 1 = on
25 N_term_M_excision = 1 # enable protein N-terminal methionine excision as a variable modification, 0 = off; 1 = on
26 max_n_variable_mod = 2 # maximum number of variable modifications
27
28 # Post-translational modifications (PTMs)
29 N_term_acetylation = 0 # enable protein N-terminal acetylation as a variable modification / score sites, 0 = off; 1 = on
30 Phosphorylation = 0 # enable S, T, Y phosphorylation as a variable modification / score sites, 0 = off; 1 = on
31 K_GG_addcut = 0 # enable -GG adduct on lysines (left after tryptic digest of an attached ubiquitin) as a variable
32
33 # Precursor ion properties
34 min_prec_charge = 1 # minimum precursor charge
35 max_prec_charge = 4 # maximum precursor charge
36 min_prec_mz = 400 # minimum precursor m/z
37 max_prec_mz = 1200 # maximum precursor m/z
38
39 # Fragment ion properties
40 min_frag_mz = 200 # minimum fragment ion m/z
41 max_frag_mz = 1800 # maximum fragment ion m/z
42
43 # Other parameters
44 n_CPU_threads = 64 # number of CPU threads used for running
45 log_level = 3 # specify how detailed the information reported in the log file, use numbers between 0-5
46 additional_options = null # null (default); others, e.g. --var-mod Dimethyl,28.0313,KR
```

Change the 'fasta_file' and 'library_name' in the 'diann_1_LibGen.params' file and then run the following command:

```
python /hpc/authorized_apps/proteomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/diann_LibGen.py
diann_1_LibGen.params
```

Appendix B. Searching and quantifying precursors: library-based search

This step will search the MS data with the predicted spectral library.

```
diann_2_Search.params
1 # DIA-NN search and quantification parameters (Version: 0.1.0, Date: 10/08/2025)
2
3 # Input: mzML or raw files (for Thermo instruments) or diaPASEF .d files (for timsTOF instruments)
4 # for multiple runs, end each file with ";"
5 input_runs = /home/zyuanli/DIATutorial_DIANN/data/hela_200ngDIA_test2_S1-B2_1_299.d;
6 /home/zyuanli/DIATutorial_DIANN/data/hela_200ngDIA_test3_S1-B3_1_300.d;
7 /home/zyuanli/DIATutorial_DIANN/data/hela_200ngDIA_test4_S1-B4_1_301.d;
8 /home/zyuanli/DIATutorial_DIANN/data/hela_200ngDIA_test5_S1-B5_1_302.d;
9
10 # Spectral library (recommend using the .speclib format file)
11 SpecLib = /home/zyuanli/DIATutorial_DIANN2/lib20241127/HUMAN/human_TryP_mc2_c0_lib/human_TryP_mc2_c0_lib.predicted.speclib
12
13 # Fasta files (use ";" to combine multiple FASTA files to one row for "fasta_file")
14 fasta_file = /home/zyuanli/DIATutorial_DIANN2/lib20241127/HUMAN/HUMAN_6_sp_tr_uc_snp_con_cu.fasta
15 re_annotate = 0 # 0 = false (default), 1 = true
16
17 # Use existing .quant files
18 use_quant = 0 # 0 = false (default), 1 = true
19 quantities_matrices = 1 # 0 = false, 1 = true (default)
20 XICs = 1 # 0 = false, 1 = true (default)
21
22
23 # Output directory name
24 outputDir = search_results_20251003
25
26 # Identification related parameters
27 ms2_tolerance = 0 # MS2 mass accuracy (ppm), use 0 for automatic inference
28 ms1_tolerance = 0 # MS1 mass accuracy (ppm), use 0 for automatic inference
29 n_scans_per_peak = 0 # number of scans per peak, use 0 for automatic inference
30 unrelated_runs = 0 # Different runs will be treated as unrelated, 0 = false (default); 1 = true
31 fdr_prec = 0.01 # FDR at precursor level
32 scoring = 2 # 1 = Generic (default); 2 = Peptidoforms; 3 = Proteoforms
33 machine_learning = 3 # 1 = Linear classifier; 2 = NNs (fast); 3 = NNs (cross-validated) (default)
34
35 # Fixed modifications
36 C_carbamidomethylation = 0 # enable cystine carbamidomethylation as a fixed modification, 0 = off; 1 = on
37
38 # Variable modifications
39 M_oxidation = 1 # enable methionine oxidation as a variable modification, 0 = off; 1 = on
40 N_term_M_excision = 1 # enable protein N-terminal methionine excision as a variable modification, 0 = off; 1 = on
41 max_n_variable_mod = 2 # maximum number of variable modifications
42
43 # Post-translational modifications (PTMs)
44 N_term_acetylation = 0 # enable protein N-terminal acetylation as a variable modification / score sites, 0 = off; 1 = on
45 Phosphorylation = 0 # enable S, T, Y phosphorylation as a variable modification / score sites, 0 = off; 1 = on
46 K_GG_addcut = 0 # enable -GG adduct on lysines (left after tryptic digest of an attached ubiquitin) as a variable modification
47
48 # Match between runs (MBR)
49 MBR = 1 # create a spectral library from DIA data and reanalyze using this library, 0 = off; 1 = on (default)
50
51 # Quantification related parameters
52 quantification_strategy = 3 # 1 = Legacy (direct); 2 = QuantUMS (high accuracy); 3 = QuantUMS (high precision) (default)
53 cross_run_normalization = 4 # 1 = Global; 2 = RT-dependent (default); 3 = RT & signal-dep. (experimental); 4 = off
54
55 # Protein inference mode
56 prot_inference = 1 # 0 = off; 1 = on (default), if turned on, one protein will be assigned to only one protein group
57 proteotypicity = 4 # 1 = Isoform IDs; 2 = Protein names (from FASTA); 3 = Genes (Species-specific); 4 = Genes (default)
58
59 # Other parameters
60 library_generation = 2 # 1 = IDs profiling; 2 = IDs, RT & IM profiling (default); 3 = Smart profiling; 4 = Full profiling
61 speed_RAM_usage = 1 # 1 = Optimal results (default); 2 = Low RAM usage; 3 = Low RAM & high speed; 4 = Ultra-fast
62 n_CPU_threads = 64 # number of CPU threads used for running
63 log_level = 5 # specify how detailed the information reported in the log file, use numbers between 0-5
64 additional_options = null # null (default); others, e.g. --var-mod Dimethyl,28.0313,KR
```

DIA-NN supports Bruker .d raw file. For Thermo data, please convert to .mzML file (see [Appendix D](#) for detail).

In the 'diann_2_Search.params' file, set the path for your raw files and spectral library and the name of output folder. Then run the following command:

```
python /hpcf/authorized_apps/proteomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/diann_Search.py
diann_2_Search.params
```

Note: Some parameters should be the same as 'diann_1_LibGen.params' (e.g. Fixed/Variable modifications).

Appendix C. Quantifying proteins: protein quantification using DIANN native or MaxLFQ algorithm

After the second step, DIA-NN will output a 'report.parquet' file that can be used for further analysis. The 'report.parquet' file mainly contains the precursor-level identification and quantification information.

```
diann_3_iqQuan.params
1 # Protein-level quantification parameters (Version: 0.1.0, Date: 10/08/2025)
2
3 # DIA-NN report file (report.parquet)
4 diann_report = /research/rgs01/home/clusterHome/zyuan1/DIATutorial_DIANN/data/search_results_20251003/report.parquet
5
6 # Protein annotation file (This can be generated by parsing the FASTA file)
7 prot_anno_file = /home/zyuan1/DIATutorial_DIANN2/lib20241127/HUMAN/human_TryP_mc2_c0_prot_anno/human_TryP_mc2_c0_prot_anno.csv
8
9 # Output directory
10 outputDir = quan_results_20251003
11
12 # Quantification method (Important!)
13 QuanMethod = 1                                # 0 = DIANN native (use PG.TopN)
14                                                # 1 = MaxLFQ (use PG.MaxLFQ)
15
16 # Q value cutoffs
17 Lib.Q.Value = 0.01                             # Please use the same column names from DIA-NN report
18 Lib.PG.Q.Value = 0.01
19
20 # Filter
21 allowed_miss_pct = 100                         # filter out proteins based on missing value percentages
22
23 # Normalization
24 perform_normalization = 1                       # 0 = off, 1 = on
25 normalization_method = median                  # use median or mean intensity for cross-run normalization
26 percentage_trimmed = 15                        # percentage of most variable intensities to be trimmed for the normalization
27 norm_species = ALL                             # if it is on, normalized to : ALL (default), 1 species for Benchmark: HUMAN;
```

In the 'diann_3_iqQuan.params' file, set the path for the 'report.parquet' file and the name of output folder. You can also choose whether to perform the cross-run normalization. Command for this step:

```
python /hpcf/authorized_apps/proteomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/diann_iqQuan.py
diann_3_iqQuan.params
```

Note: At this step you also need to input a 'protein annotation file' to add extra protein annotations in the final reports. The protein annotation file can be generated by parsing the FASTA file. Ideally, it should be the same file you used to generate the spectral library. If you use the pre-built libraries, you can use the annotation files that are already generated. If you used a customized FASTA file to build library, you need to generate the protein annotation file using the following command.

```
python /hpcf/authorized_apps/proteomics_apps/DIANN_CMD/diann_iq/diann2.2.0/20251008/parse_FASTA.py output_folder_name
file1.fasta file2.fasta ...
```

Appendix D. Thermo data conversion

MSConvertGUI is used for converting .raw file to mzML file. The software can be downloaded here (<https://proteowizard.sourceforge.io/download.html>).

DIA-NN requires using the following settings for the conversion.

The screenshot shows the MSConvertGUI (64-bit) window. The 'List of Files' radio button is selected. The 'File' field is empty, with 'Browse' and 'Add' buttons. The 'Browse network resource...' dropdown is set to 'Vendor (does not work for UNIFI, and it MUST be the first filter!)'. The 'About MSConvert' button is in the top right. The 'Filters' section shows 'Peak Picking' selected. The 'Algorithm' dropdown is set to 'Vendor (does not work for UNIFI, and it MUST be the first filter!)'. The 'MS Levels' field is set to '1'. The 'Min SNR' field is set to '0.1'. The 'Min peak spacing' field is set to '0.1'. The 'Output Directory' field is empty, with a tooltip that says 'Useful options for controlling output format and file size.' The 'Options' section shows 'Output format' set to 'mzML', 'Extension' set to 'mzML', 'Binary encoding precision' set to '32-bit', 'Write index' checked, 'Use zlib compression' unchecked, 'TPP compatibility' unchecked, 'Package in gzip' unchecked, 'Use numpress linear compression' unchecked, 'Use numpress short logged float compression' unchecked, 'Use numpress positive integer compression' unchecked, 'Combine ion mobility scans' unchecked, 'SIM as spectra' unchecked, and 'SRM as spectra' unchecked. The 'Presets' dropdown is set to 'Generic Defaults', with a 'Save Preset' button. The 'Files to convert in parallel' field is set to '1', with a 'Start' button. A table of filters is shown below the 'Options' section:

Filter	Parameters
peakPicking	vendor msLevel=1-
titleMaker	<RunId>.<ScanNumber>.<ScanNumber>.<ChargeState> File:"<SourcePath>", Nati...